

AI-Powered Customer Support Ticketing System

Core Domain: Salesforce Administrator Using AI Agentforce

Group Members-

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2. PROJECT DESCRIPTION

System Overview

The **AI-Powered Customer Support Ticketing System** is an intelligent cloud-based platform designed to improve the efficiency and effectiveness of customer support services. The system allows customers to raise service requests or complaints in the form of support tickets, which are then managed and resolved by support agents through a structured and automated workflow.

The system integrates **Artificial Intelligence (AI)** to analyze customer queries, categorize issues automatically, and assist support agents by providing relevant suggestions for resolving problems. By using AI-based analysis, the system reduces manual effort and helps support teams respond to customer queries more quickly and accurately.

The platform is developed using **Salesforce**, a widely used cloud CRM platform that enables organizations to build scalable applications, automate workflows, and manage customer interactions efficiently. The system also uses **GitHub** for project repository management and **SkillWallet** for task allocation and project tracking.

Overall, the system provides a centralized environment where customer issues can be logged, monitored, analyzed, and resolved in a structured manner.

Problem Domain

In today's digital business environment, companies receive a large volume of customer support requests every day. These requests may include technical issues, billing queries, account problems, product complaints, or general inquiries. Managing such a high number of requests manually can lead to several challenges, such as delayed responses, improper categorization of issues, and inefficient tracking of ticket status.

Traditional customer support systems often rely heavily on manual processes, where support agents must read and categorize each ticket individually. This process can be time-consuming and may result in errors or delays, especially when the number of support requests increases significantly.

The **AI-Powered Customer Support Ticketing System** addresses these challenges by introducing automation and AI-driven decision-making. The system automatically categorizes support tickets, analyzes customer sentiment, and suggests possible solutions. Additionally, the system applies **Service Level Agreement (SLA) policies** to ensure that customer issues are resolved within predefined time limits, thereby improving service quality and customer satisfaction.

Core Technology Used

The development of this system involves several modern technologies that support cloud computing, automation, and intelligent decision-making.

The primary platform used for development is **Salesforce**, which provides a powerful environment for creating custom objects, workflows, and automation rules. Salesforce allows developers to build scalable applications that can manage large volumes of data while maintaining high performance and security.

Artificial Intelligence capabilities are integrated into the system to analyze support tickets and provide intelligent feedback. AI can evaluate the content of customer queries, determine their sentiment (positive, neutral, or negative), and recommend actions for support agents.

Version control and collaboration for the project are managed using **GitHub**, which stores the project repository and documentation. Additionally, the project tasks, instructions, and learning modules are managed through **SkillWallet**, which serves as the platform for project allocation and progress tracking.

Key Capabilities

The AI-Powered Customer Support Ticketing System provides several advanced features that enhance the overall customer support process.

One of the primary capabilities of the system is **ticket creation and management**, which allows customers to submit support requests easily. Once a ticket is created, the system stores all relevant details such as the issue description, category, and priority level.

Another important capability is **automatic categorization of support issues**. The system organizes tickets into predefined categories such as technical issues, billing problems, or

account-related queries. This classification helps support agents quickly identify the type of problem and assign it to the appropriate department.

The system also provides **AI-based feedback and analysis**, which examines the content of customer messages and generates suggestions that may help resolve the issue faster. This reduces the time agents spend analyzing the problem manually.

In addition, the system incorporates **Service Level Agreement (SLA) management**, which defines response and resolution deadlines for different types of support tickets. This ensures that high-priority issues are handled promptly and that customers receive timely assistance.

Finally, the system provides **ticket tracking and monitoring**, allowing support agents and administrators to monitor ticket status, resolution progress, and overall support performance.

Target Users

The AI-Powered Customer Support Ticketing System is designed to serve multiple types of users who interact with the customer support process.

The primary users are **customers**, who use the system to submit support requests and track the progress of their issues. Customers benefit from faster responses and improved service quality.

Another important group of users is **customer support agents**, who manage and resolve support tickets. The system helps agents by automatically organizing tickets, providing AI-based insights, and ensuring that support requests are handled efficiently.

The system is also used by **administrators or system managers**, who configure support categories, define SLA policies, and monitor the overall performance of the support system.

Real-world Relevance

Customer support systems are essential for maintaining strong relationships between organizations and their customers. Businesses across industries such as e-commerce, banking, telecommunications, healthcare, and information technology rely heavily on efficient support systems to address customer concerns.

An AI-powered ticketing system significantly improves the quality of customer service by automating routine tasks, reducing response time, and providing intelligent recommendations to support agents. This leads to better customer satisfaction, improved operational efficiency, and stronger brand reputation.

In real-world applications, similar systems are widely used by large companies to manage thousands of customer requests every day. By implementing AI-based automation and cloud-

based infrastructure, organizations can scale their support operations while maintaining consistent service quality.

3. APPLICATION SCENARIOS / USE CASES

1. Enterprise Usage

Large organizations often receive thousands of customer queries every day through emails, websites, or service portals. Managing these queries manually can be inefficient and time-consuming. The **AI-Powered Customer Support Ticketing System** helps enterprises automate the process of handling customer issues.

When a customer submits a request, the system automatically creates a ticket and categorizes the issue based on its content. Artificial Intelligence analyzes the message, identifies the type of problem, and suggests possible solutions to support agents. SLA policies ensure that critical issues are prioritized and resolved within a specific time frame. This allows enterprises to improve customer satisfaction, reduce response time, and increase operational efficiency.

2. End-User Usage

From the customer's perspective, the system provides a convenient and structured way to report problems or ask for assistance. Instead of sending multiple emails or waiting on phone calls, customers can simply submit a support ticket through the platform.

Once a ticket is created, customers can track its progress, receive updates, and communicate with support agents until the issue is resolved. The AI system ensures that tickets are routed to the appropriate support team, which helps customers receive faster and more accurate solutions.

3. Educational Usage

Educational institutions such as universities and colleges can use this system to manage student support services. Students often need assistance with issues related to course registration, technical problems with learning platforms, examination queries, or administrative services.

By implementing an AI-powered ticketing system, institutions can automatically categorize student requests and assign them to the appropriate department. This reduces administrative workload and ensures that student concerns are addressed efficiently.

4. Industrial Usage

In industrial environments, companies often deal with product support, technical maintenance requests, and service complaints. The AI-powered ticketing system can be used to manage product-related issues reported by customers or internal staff.

For example, if a customer reports a malfunction in a product, the system can automatically categorize the issue and forward it to the technical support team. The AI analysis can also detect patterns in recurring problems, helping organizations identify common defects and improve product quality.

4. TECHNICAL ARCHITECTURE OVERVIEW

High-Level Architecture Diagram

The system architecture is built on the **Salesforce cloud platform**, which provides tools for managing data objects, workflows, and automation. The architecture consists of three main layers: the user interface layer, the application logic layer, and the data storage layer.

The user interface allows customers and support agents to interact with the system by creating and managing support tickets. The application layer processes requests, performs AI-based analysis, and enforces SLA policies. The data layer stores information related to tickets, support categories, AI feedback, and service policies.

Customer / Support Agent



User Interface (Ticket Portal)



Application Logic

- Ticket Processing
- AI Analysis
- SLA Management



Salesforce Platform



Database Objects

- Support Category

- AI Feedback
- SLA Policy

Component Interaction Diagram

The system consists of multiple components that interact with each other to manage support requests effectively.

- **Ticket Creation Module:** Allows customers to submit support requests.
- **Support Category Module:** Classifies tickets based on the issue type.
- **AI Feedback Module:** Analyzes ticket content and generates suggestions.
- **SLA Policy Module:** Determines response and resolution deadlines.
- **Support Agent Dashboard:** Allows agents to manage and resolve tickets.

Interaction Flow:

Customer submits ticket



Ticket Creation Module



Support Category Object assigns category



AI Feedback Module analyzes ticket



SLA Policy determines priority and deadline



Support Agent receives and resolves ticket

Data / Request Flow

The system follows a structured data flow to ensure that customer requests are handled efficiently.

1. The customer submits a support request through the system interface.

2. The system creates a new ticket and stores it in the database.
3. The ticket is automatically categorized using predefined support categories.
4. AI analyzes the ticket content and generates feedback or solution suggestions.
5. The SLA policy module determines the priority level and response deadline.
6. The ticket is assigned to a support agent for resolution.
7. The support agent updates the ticket status after resolving the issue.
8. The customer receives confirmation and updates regarding the resolution.

This workflow ensures that every request is processed systematically, reducing delays and improving overall support efficiency.

Technical Architecture Layers

1. Frontend Layer

The **Frontend Layer** represents the user interface of the system where users interact with the application. It provides the visual components that allow customers and support agents to create, view, and manage support tickets.

In this project, the frontend layer is developed using the user interface capabilities provided by **Salesforce**, which allows the creation of interactive dashboards, forms, and record pages without extensive manual coding.

Through the frontend interface, customers can submit support requests by filling out a ticket form that includes information such as the issue description, category, and priority. Support agents can access dashboards where they can view assigned tickets, track ticket status, update resolutions, and communicate with customers.

The frontend layer focuses on providing a **user-friendly and responsive interface** so that users can easily interact with the system. It also ensures that information is displayed clearly, helping support agents quickly understand customer issues and take appropriate actions.

2. Backend Layer

The **Backend Layer** is responsible for processing the logic and operations that occur behind the scenes in the system. It manages the application logic, workflows, automation rules, and data processing required to handle support tickets efficiently.

In the AI-Powered Customer Support Ticketing System, the backend is powered by **Salesforce**, which provides tools for building automation workflows, triggers, and custom objects that manage system functionality.

The backend performs several important tasks, including:

- Processing newly created support tickets
- Automatically categorizing tickets into predefined support categories
- Triggering AI analysis to generate feedback and suggestions
- Applying SLA policies to determine response and resolution deadlines
- Assigning tickets to appropriate support agents

By automating these processes, the backend layer ensures that customer requests are handled quickly and efficiently without requiring manual intervention for every step.

3. Database Layer

The **Database Layer** is responsible for storing and managing all the information used by the system. This includes data related to customer tickets, support categories, AI analysis results, and SLA policies.

In this project, the database layer is managed within **Salesforce**, which provides a secure cloud-based data storage system using custom objects and fields.

Key database objects used in the system include:

- **Support Category Object** – Stores different categories of support issues such as technical problems, billing queries, or account issues.
- **AI Feedback Object** – Stores the analysis results generated by the AI system, including sentiment and suggested solutions.
- **SLA Policy Object** – Stores predefined service policies that determine response times and resolution deadlines for tickets.

The database layer ensures that all system data is stored securely and can be retrieved quickly whenever needed.

4. External APIs / Services

External APIs and services enhance the functionality of the system by allowing integration with additional technologies such as Artificial Intelligence tools or third-party services.

In this project, AI capabilities may be integrated through external services or built-in AI features provided by **Salesforce**. These AI services analyze customer messages and provide insights such as sentiment analysis, recommended actions, and automated categorization.

External services may also include communication tools that send notifications or updates to customers when the status of their ticket changes. These integrations help the system provide a more intelligent and automated support experience.

5. Deployment Environment

The **Deployment Environment** refers to the platform where the system is hosted and executed. In this project, the system is deployed on the **cloud infrastructure provided by Salesforce**.

Cloud deployment offers several advantages, including scalability, reliability, and accessibility. Since the system is cloud-based, users can access it from any location using a web browser without installing additional software.

The deployment environment ensures that the system remains secure, continuously available, and capable of handling large volumes of customer support requests.

5. PREREQUISITES

5.1 Software Requirements

To successfully design, develop, and deploy the **AI-Powered Customer Support Ticketing System**, several software tools and platforms are required. These tools help in development, testing, version control, and deployment of the application.

Development Tools

Development tools are required to design, configure, and manage different components of the system. The primary development platform used in this project is **Salesforce**, which provides tools for creating custom objects, managing workflows, and implementing automation features.

The system uses the Salesforce platform to configure objects such as **Support Category**, **AI Feedback**, and **SLA Policy**, which are essential for managing support tickets and system operations. Salesforce also provides tools for managing data, creating reports, and configuring user interfaces.

For collaboration and version control, the project repository is maintained on **GitHub**, which allows developers to store project files, track changes, and collaborate efficiently.

IDE (Integrated Development Environment)

An Integrated Development Environment (IDE) is used to write, edit, and manage project code and configuration files. For this project, **Visual Studio Code** is commonly used as the development environment.

Visual Studio Code provides useful features such as syntax highlighting, debugging tools, extensions, and integration with version control systems like GitHub. It helps developers efficiently manage project files and make configuration changes when necessary.

Runtime Environment

The runtime environment refers to the platform where the application executes its processes and operations. In this project, the runtime environment is provided by **Salesforce**, which manages the execution of workflows, triggers, and automation processes related to ticket management.

Salesforce handles the backend processing of support tickets, AI analysis, and SLA policy enforcement, ensuring that the system runs smoothly without requiring additional server management.

Cloud Services

The project is deployed on the cloud infrastructure provided by **Salesforce**, which offers secure and scalable cloud services for hosting the application.

Cloud deployment allows users to access the system from any location using a web browser, making the platform highly accessible and reliable. The cloud environment also provides automatic data backup, security features, and scalability to support a large number of customer support requests.

Additionally, project management and learning resources are accessed through **SkillWallet**, which provides task tracking and project instructions.

5.2 Libraries / Frameworks / Dependencies

The development of the **AI-Powered Customer Support Ticketing System** relies on several libraries, frameworks, and platform features that support application development, automation,

and data management. These components help in building a scalable and efficient customer support system.

1. Salesforce Lightning Framework

The primary framework used for building the user interface and application logic is the **Salesforce Lightning** framework. It provides a modern component-based architecture that allows developers to create interactive and responsive applications on the **Salesforce** platform.

Lightning framework enables the creation of custom components, dashboards, and record pages that support ticket creation, ticket tracking, and support agent workflows.

2. Apex Programming Language

Salesforce Apex is a strongly typed programming language used within the **Salesforce** platform to implement backend logic. Apex allows developers to create custom triggers, classes, and business logic that automate system processes.

In this project, Apex can be used to:

- Process ticket creation events
 - Trigger AI analysis when a ticket is submitted
 - Enforce SLA policies
 - Automate ticket assignment to support agents
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3. Salesforce Objects and Data Model

The system relies on **custom objects** and **standard objects** provided by **Salesforce** for data storage and management.

Important custom objects include:

- **Support Category Object** – stores different categories of support issues
- **AI Feedback Object** – stores AI-generated insights and suggestions
- **SLA Policy Object** – stores service level agreement policies

These objects act as the **data model** of the application and allow the system to store, retrieve, and manage support-related information.

4. REST APIs

The system may use **REST APIs** to integrate external services or AI tools with the **Salesforce** platform. REST APIs enable communication between different software systems by allowing them to exchange data through HTTP requests.

REST APIs can be used for:

- AI-based ticket analysis
 - Integration with external customer service platforms
 - Sending notifications or updates to users
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5. Git Version Control

Version control is managed using **Git** and **GitHub**. These tools help track changes in project files, manage different versions of the code, and enable collaboration among developers working on the project.

Using version control ensures that project files remain organized and allows developers to revert to previous versions if necessary

5.3 Hardware Requirements

The **AI-Powered Customer Support Ticketing System** is a cloud-based application developed and deployed on the **Salesforce** platform. Since the system operates on cloud infrastructure, it does not require specialized hardware components for deployment. However, certain basic hardware specifications are required for users and administrators to access and manage the system effectively.

System Specifications

To develop, access, and operate the system smoothly, a computer system with standard specifications is required. These specifications ensure that users can run development tools, access cloud services, and manage the application efficiently.

Recommended system specifications include:

- **Processor:** Intel Core i3 or higher / equivalent processor
- **RAM:** Minimum 4 GB (8 GB recommended for development tasks)

- **Storage:** At least 20 GB of available disk space
- **Operating System:** Windows 10/11, macOS, or Linux
- **Internet Connection:** Stable broadband internet connection

These specifications allow developers to run development environments such as **Visual Studio Code** and access cloud platforms like **Salesforce** and **GitHub** without performance issues.

Devices / Sensors

The **AI-Powered Customer Support Ticketing System** does not require specialized hardware devices or sensors because it is primarily a software-based application. The system operates through web browsers and cloud services, which eliminates the need for additional physical devices.

However, users typically access the system using standard computing devices such as:

- Desktop computers
- Laptop computers
- Tablets or mobile devices with web browsers

These devices allow customers and support agents to interact with the system, submit support tickets, monitor ticket status, and manage support operations.

6. PRIOR KNOWLEDGE REQUIRED

Before implementing the **AI-Powered Customer Support Ticketing System**, certain foundational technical concepts are required. These concepts help developers understand how the system is designed, how different components interact with each other, and how the platform processes and manages data efficiently.

Programming Language Knowledge

Basic knowledge of programming languages is important for developing and customizing the system. In this project, backend logic and automation within **Salesforce** can be implemented using **Salesforce Apex**, which is a strongly typed object-oriented programming language similar to Java.

Understanding programming concepts such as variables, conditional statements, loops, functions, and object-oriented principles helps developers create automation rules, triggers, and custom logic required for ticket management and AI integration.

Additionally, knowledge of general programming languages like Python, Java, or JavaScript can be helpful when integrating external services or APIs.

Framework Basics

Developers should have a basic understanding of application frameworks, especially the **Salesforce Lightning** framework used for building user interfaces on the **Salesforce** platform.

The Lightning framework provides a component-based architecture that allows developers to design responsive and interactive interfaces for ticket management dashboards, support forms, and administrative panels. Understanding how components interact and how UI elements are structured is important for building an efficient and user-friendly interface.

Database Fundamentals

Since the system stores and processes a large amount of support-related data, knowledge of database fundamentals is essential. Developers should understand concepts such as:

- Data tables or objects
- Fields and attributes
- Relationships between objects
- Data storage and retrieval
- Basic querying techniques

In this project, **Salesforce** uses **custom objects and fields** to manage data such as support categories, AI feedback results, and SLA policies. Understanding how these objects are structured and related helps ensure efficient data management within the system.

Networking / Cloud Concepts

Because the system is deployed on a cloud platform, knowledge of basic networking and cloud computing concepts is important. Developers should understand how cloud platforms host applications, manage user access, and ensure secure communication between different system components.

The system is hosted on the **cloud infrastructure of Salesforce**, which allows users to access the application from any location through a web browser. Understanding concepts such as cloud deployment, client-server architecture, and web-based services helps developers design scalable and reliable systems.

AI/ML Fundamentals (If Applicable)

Since the system incorporates Artificial Intelligence for analyzing customer support tickets, a basic understanding of AI and machine learning concepts is beneficial.

AI technologies can be used to analyze customer messages, detect sentiment, identify keywords, and suggest possible solutions to support agents. These capabilities help automate ticket categorization and improve response time.

While deep knowledge of machine learning algorithms may not be required for this project, understanding fundamental AI concepts such as text analysis, pattern recognition, and predictive recommendations can help developers integrate intelligent features into the system.

7. PROJECT OBJECTIVES

The main objective of the **AI-Powered Customer Support Ticketing System** is to design and implement an intelligent platform that improves the efficiency and effectiveness of customer support operations. The system aims to automate ticket management, provide AI-based insights, and ensure timely resolution of customer issues through structured workflows and service policies.

The objectives of the project can be categorized into technical objectives, performance objectives, deployment objectives, and learning outcomes.

Technical Objectives

The technical objective of this project is to design and develop a scalable customer support management system using modern cloud-based technologies. The system should be capable of creating, managing, and tracking support tickets efficiently.

Another key objective is to implement custom data structures and objects within **Salesforce**, such as Support Category, AI Feedback, and SLA Policy objects, to organize and store system data effectively. The project also aims to integrate Artificial Intelligence features that can analyze customer queries, categorize issues automatically, and provide suggestions to support agents.

Additionally, the system should provide an intuitive user interface for both customers and support agents, allowing them to interact with the platform easily. Proper data management, automation workflows, and secure data handling are also part of the technical objectives.

Performance Objectives

The performance objective of the system is to improve the speed and efficiency of handling customer support requests. The system should reduce the time required to categorize and process customer tickets by automating repetitive tasks using AI.

Another performance goal is to ensure quick response and resolution of customer issues through the implementation of Service Level Agreement (SLA) policies. These policies define response times and resolution deadlines, ensuring that high-priority issues are addressed promptly.

The system should also be capable of handling a large number of support requests without significant delays, ensuring smooth performance even during high demand.

Deployment Objectives

The deployment objective of the project is to host and run the application on a secure and scalable cloud platform. The system is deployed using the cloud infrastructure provided by **Salesforce**, which allows users to access the application from any location using a web browser.

Cloud deployment ensures high availability, data security, and easy system maintenance. It also allows organizations to scale their support operations without the need for additional hardware infrastructure.

The project also aims to maintain proper project documentation and version control through platforms such as **GitHub**, ensuring organized development and collaboration.

Learning Outcomes

This project provides valuable learning opportunities for understanding modern cloud-based application development and AI integration. Through this project, developers gain practical experience in working with the **Salesforce** platform, including creating custom objects, managing workflows, and configuring automation rules.

Participants also develop skills in system design, database management, and project documentation. The project helps learners understand how Artificial Intelligence can be integrated into real-world business applications to improve efficiency and decision-making.

Overall, the project enhances knowledge of cloud computing, customer relationship management systems, and intelligent automation technologies.

8. SYSTEM WORKFLOW

The AI-Powered Customer Support Ticketing System follows a structured workflow that ensures customer issues are processed efficiently and resolved within defined service policies.

Step 1: User Interaction

The workflow begins when a customer accesses the support portal through the system interface. The customer fills out a support ticket form containing details such as the issue description, category (if known), and priority level.

Step 2: Input Handling

Once the customer submits the ticket, the system validates the input data to ensure that all required fields are completed. The ticket information is then stored in the Salesforce database as a new ticket record.

Step 3: Processing Logic

After the ticket is created, the backend system processes the request. The ticket is analyzed using predefined rules and AI-based processing to determine the appropriate category and urgency of the issue.

Step 4: Core System Execution

The AI module analyzes the content of the customer query. It performs operations such as:

- Keyword detection
- Sentiment analysis
- Issue classification

Based on this analysis, the system categorizes the ticket and determines the most suitable support team.

Step 5: Output Generation

The system generates AI-based feedback and suggested solutions that can assist support agents in resolving the issue. SLA policies are applied to determine response deadlines and resolution timelines.

Step 6: Response Delivery

The support agent receives the assigned ticket through the agent dashboard. After resolving the issue, the agent updates the ticket status. The system then notifies the customer and provides resolution details.

This structured workflow ensures that customer support requests are handled efficiently, reducing response time and improving service quality.

9. MILESTONE 1: REQUIREMENT ANALYSIS & SYSTEM DESIGN

9.1 Problem Definition

Organizations receive a large number of customer support requests every day. Handling these requests manually can lead to delays, misclassification of issues, and inefficient ticket tracking.

Traditional systems often require support agents to manually read and categorize each ticket, which increases workload and response time. This can negatively impact customer satisfaction.

The AI-Powered Customer Support Ticketing System addresses these challenges by automating ticket classification, applying SLA policies, and providing AI-based recommendations for faster issue resolution.

9.2 Functional Requirements

The system must support the following functionalities:

- Customers must be able to create and submit support tickets.
 - The system must store ticket details in the database.
 - The system must automatically categorize tickets.
 - AI should analyze ticket content and generate suggestions.
 - The system must apply SLA policies to determine response deadlines.
 - Support agents must be able to view, update, and resolve tickets.
 - Administrators must manage support categories and SLA policies.
 - Customers must be able to track ticket status.
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9.3 Non-Functional Requirements

The system must meet several quality and performance requirements.

Performance

- The system should process support requests quickly without delays.

Scalability

- The system should handle a large number of tickets simultaneously.

Security

- Customer data must be stored securely with controlled access.

Reliability

- The system should operate continuously with minimal downtime.

Usability

- The interface should be simple and user-friendly for customers and support agents.
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9.4 System Design Decisions

Several design decisions were made during the development of the system:

- The application is built on the **Salesforce cloud platform** for scalability and reliability.
 - Custom objects are used to store support-related data.
 - AI features are integrated to analyze ticket content and assist support agents.
 - SLA policies are implemented to ensure timely resolution of issues.
 - The system follows a **three-layer architecture (Frontend, Backend, Database)** to maintain modularity and maintainability.
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9.5 Technology Stack Selection Justification

The technologies used in this project were selected based on their reliability, scalability, and compatibility with cloud-based applications.

Salesforce Platform

- Provides a powerful CRM platform for building scalable cloud applications.

Apex Programming Language

- Used for implementing backend logic and automation workflows.

Salesforce Lightning Framework

- Used to create modern and responsive user interfaces.

GitHub

- Used for version control and project repository management.

REST APIs

- Used for integrating external AI services and communication systems.

These technologies ensure that the system remains scalable, secure, and efficient.

10. MILESTONE 2: ENVIRONMENT SETUP & INITIAL CONFIGURATION

10.1 Development Environment Setup

The development environment was prepared to support the creation and management of the AI-Powered Customer Support Ticketing System.

The following tools were configured:

- Salesforce Developer Account
- Visual Studio Code
- Salesforce CLI
- GitHub repository for version control

Developers access the Salesforce platform to create custom objects, workflows, and automation rules required for ticket management.

10.2 Dependency Installation

Several dependencies were installed to enable development and integration.

Typical setup commands include:

Install Salesforce CLI

Install VS Code Salesforce Extensions

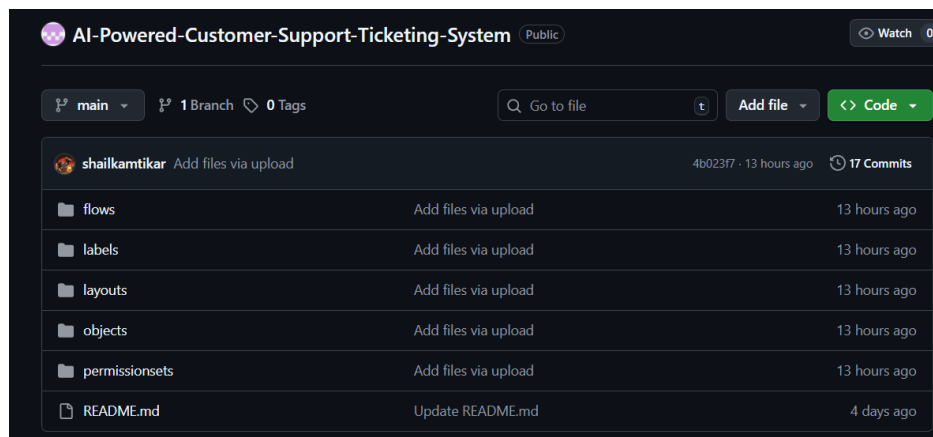
Initialize Git repository

Connect Salesforce Org

These dependencies allow developers to manage Salesforce metadata, automate workflows, and deploy project components.

10.3 Project Structure Creation

The project structure was organized to maintain clear separation between components.



This structure helps maintain code readability and simplifies future development.

flows

Contains Salesforce Flow metadata. Flows automate business processes without code. They can create or update records, send notifications, assign tickets, or guide users through steps. In this project, flows likely automate ticket creation, assignment, or status updates.

labels

Stores Custom Labels used across the application. Labels hold reusable text values such as messages, button text, or error messages. They help maintain consistency and support multi-language translations.

layouts

Defines the page layout for objects. Layouts control how fields, sections, related lists, and buttons appear on record pages in Salesforce. In a support ticket system, layouts determine how ticket details like status, priority, and customer information are displayed to agents.

objects

Contains the definitions of custom objects used in the system. Objects represent data structures similar to database tables. For example, a “Support Ticket” object might store fields like Ticket ID, Issue Description, Status, Priority, and Assigned Agent.

permissionsets

Stores Permission Set metadata. Permission sets define what users are allowed to do, such as viewing, creating, editing, or deleting records. In this project, they likely control what support agents, admins, or users can access in the ticketing system.

README.md

A documentation file that explains the project. It usually includes the project overview, purpose,

setup instructions, and details about how the AI-powered customer support ticketing system works.

10.4 Configuration Setup

System configuration includes setting up the required objects, workflows, and policies.

Key configurations include:

- Creating **Support Category Object**
- Creating **AI Feedback Object**
- Creating **SLA Policy Object**
- Setting up automation rules for ticket assignment
- Configuring user roles and permissions

These configurations ensure that the system operates efficiently and securely.

11. MILESTONE 3: CORE SYSTEM DEVELOPMENT

11.1 Feature 1 – Ticket Creation and Management

Description

This feature allows customers to create support tickets by submitting issue details through the system interface.

Implementation

The system captures customer information and issue description and stores it in the Salesforce database as a ticket record.

Output

- Ticket ID generated
- Ticket stored in database
- Notification sent to support team

	Case Number	Subject	Status	Date/Time Opened	Case Owner...
1	00001001	Performance inadequate for second consecutive week	Closed	3/4/2026, 4:31 AM	OEPIC
2	00001034		Working	3/7/2026, 1:58 PM	sha
3	00001035	Log in issue...	New	3/7/2026, 2:38 PM	sha
4	00001033		Working	3/7/2026, 1:47 PM	sha

Home page of ticketing system

11.2 Feature 2 – AI-Based Ticket Analysis

Description

Artificial Intelligence is used to analyze the content of customer tickets and provide intelligent insights.

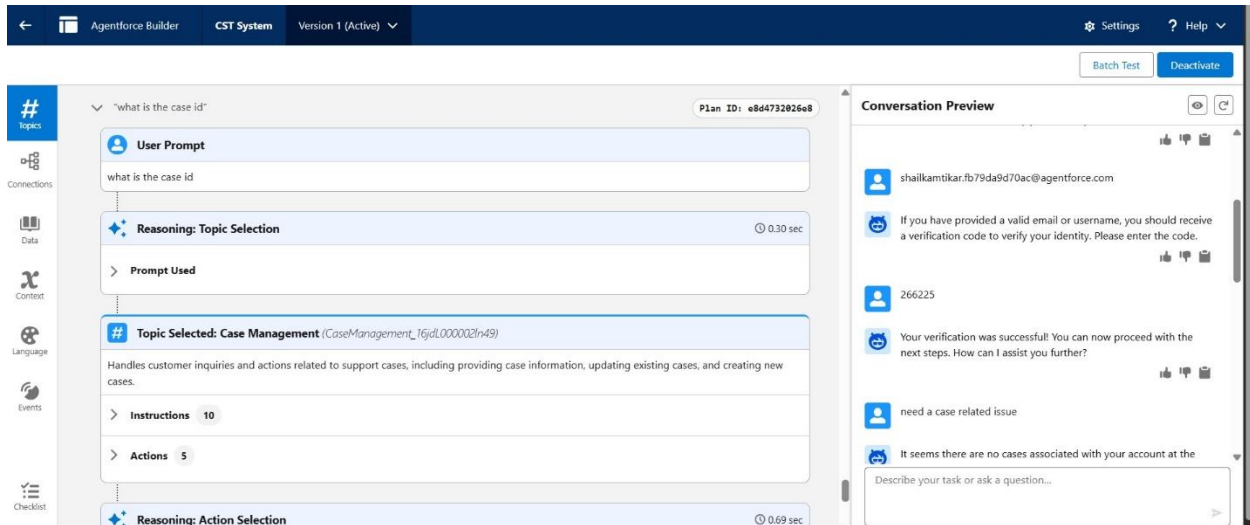
Implementation

The AI module performs tasks such as:

- Sentiment analysis
- Keyword extraction
- Issue categorization
- Solution recommendation

Output

- AI feedback stored in database
- Suggestions provided to support agents



Message Generation Flow

11.3 Feature 3 – SLA Management and Ticket Assignment

Description

The system enforces SLA policies to ensure timely resolution of support tickets.

Implementation

Based on ticket category and priority:

- Response time is calculated
- Deadline is assigned
- Ticket is automatically assigned to an available support agent

Output

- Ticket assigned to support team
- SLA deadline displayed on agent dashboard



It seems there are no cases associated with your account at the moment. Would you like me to create a new case for you? If so, please provide some details about the issue you're experiencing.





yes sure



Could you please provide a brief description of the issue you're facing? This will help me create a case with all the necessary details for you.





log in issue...



Would you like me to create a case for your log-in issue and forward it to a service representative for further assistance?



Case creation box

12. MILESTONE 4: INTEGRATION & OPTIMIZATION

12.1 Component Integration

All system components were integrated to ensure smooth communication between modules.

Integrated components include:

- Ticket Creation Module
- AI Feedback Module
- SLA Policy Module
- Support Agent Dashboard

These components work together to process customer requests automatically.

12.2 Performance Optimization

Performance optimization techniques were implemented to improve system efficiency.

These include:

- Efficient database queries
- Automated workflows
- Reduced manual intervention
- Optimized data retrieval from Salesforce objects

These optimizations help the system process large volumes of tickets quickly.

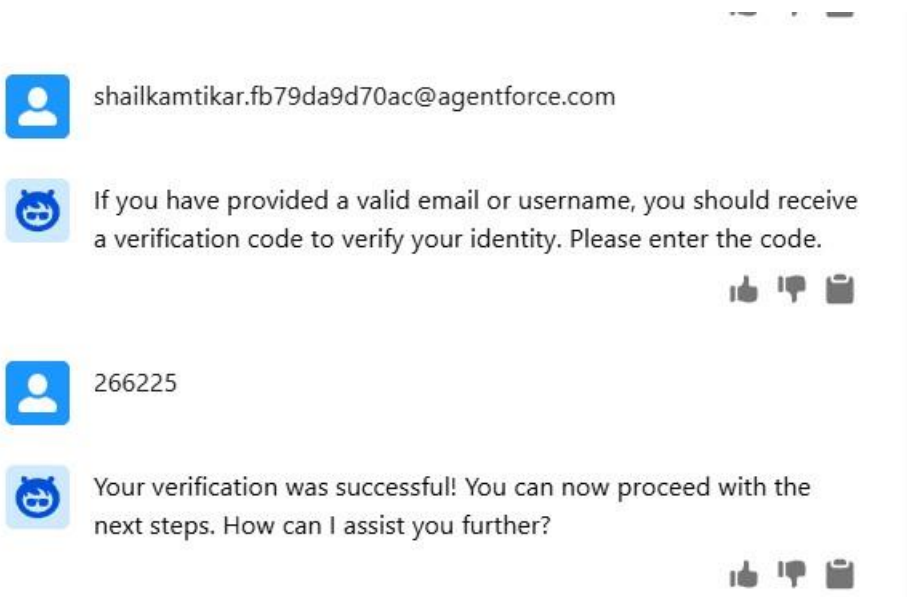
12.3 Security Enhancements

Security measures were implemented to protect system data.

Key security features include:

- User authentication and role-based access control
- Secure data storage within Salesforce
- Controlled access to sensitive customer information

These measures ensure that only authorized users can access system data.



Secure authentication via email

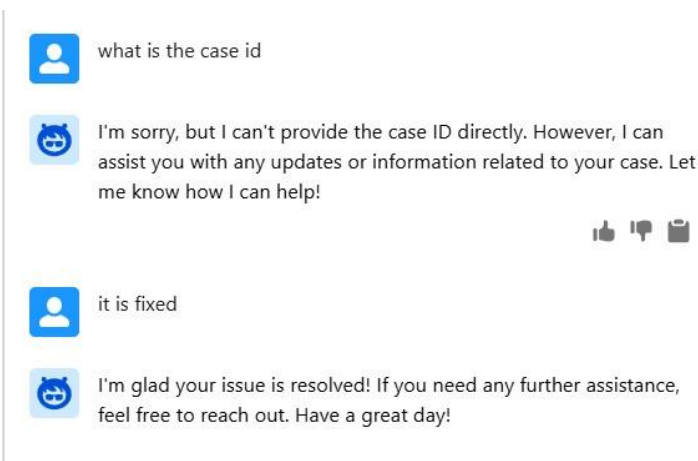
12.4 Error Handling & Validation

Error handling mechanisms were implemented to manage unexpected situations.

Examples include:

- Input validation during ticket submission
- Handling missing or incomplete data
- System alerts for failed operations
- Logging system errors for debugging

These mechanisms improve system reliability and prevent incorrect data processing.



13. MILESTONE 5: TESTING & VALIDATION

13.1 Test Cases

Test Case ID	Input	Expected Output	Status
TC01	Submit support ticket	Ticket created successfully	Passed
TC02	AI analyzes ticket	AI feedback generated	Passed
TC03	Apply SLA policy	Deadline assigned	Passed
TC04	Agent resolves ticket	Status updated to resolved	Passed

13.2 Unit Testing

Unit testing was performed on individual modules such as ticket creation, AI analysis, and SLA policy processing. Each component was tested independently to ensure proper functionality.

13.3 Integration Testing

Integration testing ensured that different modules of the system worked together correctly. This included testing interactions between ticket creation, AI analysis, and ticket assignment modules.

13.4 User Acceptance Testing

User acceptance testing was conducted to ensure the system met the needs of customers and support agents. Test users verified that tickets could be created, tracked, and resolved efficiently.

13.5 Performance Metrics

Performance metrics were evaluated to measure system effectiveness.

Metrics include:

- Ticket response time
- Ticket resolution time
- System uptime
- AI categorization accuracy

These metrics confirm that the system performs efficiently under normal usage conditions.

14. DEPLOYMENT

14.1 Deployment Architecture

The system is deployed on the Salesforce cloud platform, which hosts the application and database in a secure cloud environment.

Architecture includes:

User → Web Interface → Salesforce Application Layer → Database Objects

14.2 Hosting Platform

The system is hosted on the **Salesforce Cloud Platform**, which provides:

- High availability
 - Secure data storage
 - Scalable infrastructure
 - Web-based accessibility
-

14.3 Deployment Steps

The deployment process includes:

1. Configure Salesforce environment
 2. Deploy custom objects and workflows
 3. Deploy Apex classes and automation rules
 4. Configure user roles and permissions
 5. Test deployed system functionality
-

14.4 Production Considerations

Before deploying the system to production, several factors must be considered:

- Data security and access control
- System performance under heavy load
- Backup and recovery strategies
- Monitoring system performance

Proper deployment planning ensures that the system remains reliable and scalable in real-world operations.